VF Asia Footwear Laboratory Testing Manual

Test Method for Specific Gravity	**Test Method #: VF-A04** **REF #:TM-R-033** **Page 1 of 3**
Date Issued: November 27, 2018 **Revision #: 2** **Reason for Revision: 1. Change dimension of test sample for porous materials to 10 cm long & 10 cm wide > 2. Increase quantity of samples to be tested to 3.**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the specific gravity (relative density) of a test material. This method is applicable for rubber, plastic and foams.

REFERENCES: ASTM D792

EQUIPMENT AND MATERIAL:

1. Balance, capable of weighing to 0.001grams.
2. Support bridge, which can be positioned over the weighing pan of the balance without contacting it, and support a 250 ml glass beaker.
3. 250 ml glass beaker.
4. Clean thin wire, for suspending the test material in water.
5. Sinker, for submerging test materials which are less dense than water.
6. Bristle brush
7. Razor or knife
8. Distilled water
9. Alcohol, for removing air bubbles from the sample.
10. Thermometer, capable of measuring to 0.5ºC.
11. Rubber, plastic, closed cell foams and porous materials of uniform thickness.

NOTE: For open cell midsole and sockliner use **VF Asia** test method **VF –A03**

SAMPLE PREPARATION:

Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0˚C Temperature and 55 ±10 % Relative Humidity.

A) For EVA-based midsoles, unit-soles and test plaques (injection molded and compression- molded).

1. Die or hand cut a sample of the test material which is at least 1 cm wide by 2 cm long., and preferably 10 mm thick. ***(Note: if 10 mm thickness not possible, a thinner sample can be used provided it is not below 5 mm).*** Cut the sample from the thickest section of the heel at least 1.0 cm from the outside edge. Skin on/Skin off designation is as specified in Appendix C-Midsole Sampling.

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2. Using a razor of sharp knife, cut a slit in the sample from one edge to the center of the test sample, once the sample has been cut, use a soft bristle brush to remove any dust particles or debris from the test sample.

B) Rubber, plastic and closed-cell sock-liner foams.

1. Die or hand cut a sample of the test material which is at least 2 cm wide by 2cm long, the sample must have a thickness of at least 1 mm for each 1gram of weight. The total weight of the sample must be 1.0-5.0 grams.
2. Using a razor or sharp knife, cut a slit in the sample from one edge to the center of the test sample. Once the sample has been cut, use a soft bristle brush to remove any dust particles of debris from the test sample.

C) Porous materials of uniform thickness (PU open cell foam, paper board, etc.)
Die or hand cut a sample of the test material which at 10 cm wide and 10 cm long.

REPETITIONS: Test three samples per material

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A) Wet Method-Rubber, plastic and closed cell foams (non-porous materials).

1. Place the support bridge over the weighing pan such that no portion of the support bridge contacts the weighing pan.
2. Place the 25 ml beaker on the support bridge with distilled water. The height of water should at least 10 mm above the tray when submerged with hook.
3. Hang the tray on the hook and ensure that it will not contact from the bottom or sides of the beaker.
4. Zero in the balance by pressing T.
5. Weigh the sample in air. When the value of weight in air appears and the figure already stable press REF. Wait till the word “LIQUID “appears.

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6. Take out the sample. To avoid bubbles dip in alcohol (for 1-2 seconds) shaking any excess alcohol from the test sample then dip again in distilled water.
7. Place the sample inside the tray and ensure that the tray must not contact any portion of the beaker.
8. Immediately (to minimize any absorption of water into the sample) check the weight by pressing the icon REF
9. Value of specific gravity appears.

Dry Method – Porous materials of uniform thickness (PU open cell foam, paper board, etc.)

This test is performed in an identical manner as "VF- A02 Weight", except that the thickness (**T**) of the material is also measured (according to "VF-A03 Thickness/Gauge") to the nearest 0.001 cm. The specific gravity is calculated as follows:

CALCULATION:

$$\text{Specific Gravity (g/cm3)} = \frac{W}{A \times T}$$

REPORT: The average of the three test samples to the nearest 0.01g/cm3. If the values vary by more than 0.02g/cm^3, a fourth test sample must be measured and the sample which deviates the most from the other two samples should be discarded.